Practice Assignment 13.1 Prompt Engineering

This practice assignment helps you apply and experiment with prompt engineering techniques covered in Week 13, including:

- Instruction-based prompting
- Role-based prompting
- Chain-of-thought prompting
- Few-shot prompting
- Creative and constraint-based prompting

Note: This is a self-learning activity. *No submission is required.* The goal is to build fluency through iteration: write a prompt, observe the output, identify what is missing or weak, and refine the prompt until the result becomes clearer, more structured, and more reliable.

Instructions

You will work through **five industry scenarios**. For each scenario, you will write and test multiple prompts using different techniques (role-based, instruction-based, few-shot, chain-of-thought, constraints, and refinement). Keep a simple log of:

- Your prompt (Version 1)
- The model output
- What was missing / unclear / risky
- Your revised prompt (Version 2 / 3)
- What improved and why

Scenario 1: Business Strategy (Consulting Use Case)

Context: A mid-sized retail company is experiencing declining footfall in physical stores but has a growing online presence. Leadership wants a 12-month omnichannel growth strategy.

Practice Tasks

1. **Role-Based Prompting**
    - Assign the model a role such as:
        - Management Consultant
        - Retail Strategy Expert

2. **Instruction-Based Prompting**
   - Clearly specify:
     - Time horizon
     - Constraints (budget, existing workforce)
     - Output format (bullets, phases, roadmap)
3. **Chain-of-Thought Prompting**
   - Ask the model to:
     - Analyze the problem
     - Identify root causes
     - Propose solutions step-by-step

Reflection: *Which prompt produced the most actionable strategy? Why?*

Scenario 2: Human Resources (Talent & L&D)

Context: An organization is facing high attrition among early-career employees (0–2 years experience). HR wants to redesign the employee onboarding and engagement program.

Practice Tasks

1. **Role-Based Prompting**
   - Use roles such as:
     - HR Business Partner
     - Organizational Psychologist
2. **Constraint-Based Prompting**
   - Include constraints like:
     - Remote/hybrid workforce
     - Limited training budget
3. **Few-Shot Prompting**
   - Provide examples of:
     - Poor onboarding practices
     - Effective onboarding practices

Reflection: *How did providing examples change the quality of recommendations?*

Scenario 3: Product Management (Tech Industry)

Context: A SaaS company is launching a new AI-powered feature. The product team needs: customer positioning, feature benefits, and risk considerations.

Practice Tasks

1. **Instruction-Based Prompting**

- Ask for:
  - Value proposition
  - Target personas
  - Key differentiators

2. **Chain-of-Thought Prompting**
   - Ask the model to:
     - Identify customer pain points
     - Map features to benefits
     - Anticipate objections
3. **Multi-Role Prompting**
   - Run the same task with different roles:
     - Product Manager
     - Sales Leader
     - End Customer

Reflection: *How did outputs differ across roles?*

Scenario 4: Marketing & Brand Communication

Context: A traditional B2B company wants to reposition its brand as innovative and AI-ready without alienating existing customers.

Practice Tasks

1. **Tone & Style Constraints**
   - Specify tone:
     - Professional
     - Visionary
     - Trust-building
2. **Instruction-Based Prompting**
   - Ask for:
     - Brand messaging
     - Website homepage copy
     - Leadership talking points
3. **Prompt Refinement**
   - Rewrite the prompt to:
     - Reduce buzzwords
     - Increase clarity

Reflection: *What changes to the prompt improved brand authenticity?*

Scenario 5: Data & Decision Support

Context: A leadership team must decide whether to expand into a new geographic market.

Practice Tasks

1. **Chain-of-Thought Prompting**
   - Ask the model to:
     - Identify evaluation criteria
     - Assess risks and opportunities
     - Provide a recommendation
2. **Assumption-Based Prompting**
   - Explicitly state assumptions:
     - Market size
     - Competition
     - Regulatory complexity
3. **Structured Output Prompting**
   - Require:
     - Tables
     - Pros & Cons
     - Final decision summary

Reflection: *Did explicit assumptions improve decision quality?*

Final Reflection

Answer these for yourself:

- Which prompting technique felt most powerful?
- Which scenarios required the most refinement?
- How would you apply these techniques in your current job role?

Key Learning Goal

Prompting is not about asking better questions once — it's about **iterative thinking**, **clarity**, and **intent**.

Expected Output

By the end of this practice, you should have:

- Multiple prompt versions for each scenario (at least 2 iterations per scenario).
- Outputs generated from different prompt types (instruction-based, role-based, few-shot, chain-of-thought, constraint-based).

- A short note on what improved after refinement (clarity, structure, usefulness, tone, reduced ambiguity).

How to Approach the Project

Treat this like a mini lab, not a writing exercise. Your goal is to learn how prompt choices change outputs through repeatable iteration. For each scenario, follow the same loop:

1. **Start with a baseline prompt (Version 1).** Keep it simple and run it once. Don't overthink the first attempt—you need an output to diagnose.
2. **Evaluate the output with a quick checklist:** Is it accurate for the context? Is it actionable? Is it structured the way you need? Did it make assumptions you didn't want? Is the tone appropriate? Note 2–3 concrete issues.
3. **Refine one thing at a time (Version 2 / 3).** Avoid changing everything at once. Improve one dimension per revision—for example: add a role, add constraints, force a table/bullets, add assumptions, or add a short example.
4. **Compare outputs side-by-side.** Paste Version 1 and Version 2 outputs next to each other and write one line: "What improved?" This is where the learning happens.

Practical tips to complete it successfully:

- **Lock the task before you prompt.** Include the context, goal, constraints, and what "good" looks like. Most weak outputs happen because the prompt didn't define success.
- **Force structure early.** Ask for phases, bullet points, headings, or a table. Structure reduces randomness and makes outputs easier to judge.
- **Use roles to control depth and tone.** "Management consultant" vs "HR business partner" will change what the model prioritises—use that intentionally.
- **Use constraints to prevent generic fluff.** Add budgets, time limits, workforce limits, and channel constraints (remote/hybrid). Constraints force trade-offs, which makes outputs more realistic.
- **When outputs feel confident-but-wrong, add assumptions.** State assumptions explicitly (market size, competition, regulatory complexity). This reduces hidden guessing.
- **For few-shot prompts, keep examples short and relevant.** One good example beats five noisy ones. Make sure examples match the exact format you want back.
- **Don't chase perfect answers.** Aim for visible improvement after refinement: clearer structure, better constraints, fewer assumptions, and more actionable steps.

- **Keep a simple prompt log.** Save prompts and outputs in a notes file or a markdown cell so you can track what changed and why.

Minimum completion standard (recommended): For each of the 5 scenarios, do at least two prompt versions (V1 baseline + V2 refined) and write one line explaining what improved in V2.

Practice Guidelines:

This week, you will use **Vocareum** to practice **prompt engineering through code-based LLM calls**, applying techniques such as *instruction-based*, *role-based*, *few-shot*, *chain-of-thought*, and *constraint-based* prompting. You will run prompts for the given business scenarios, compare outputs across techniques, and refine your prompts iteratively based on output quality. You can also utilise the in-built GenAI assistant within Vocareum for assistance with prompts, debugging, and API-backed tasks. No submission is required for this activity; the focus is on fluency with prompt techniques and iterative improvement.

To Use Chat Support for This Assignment (inside Vocareum):

- Open your Week 13 practice notebook in **Vocareum**. Use the left sidebar to open the **Assistant** panel.
- Ask for clarifications, debugging help, or code snippets directly in the **Assistant** chat while you implement and test different prompting techniques.
- Monitor your **GenAI Allowance/Budget, as shown in the Assistant panel,** and plan prompts accordingly.
- If you exhaust your allowance for a task, please contact the Emeritus support team.
- **API key access:** Your course-specific API key and usage notes are provided via the guide and Vocareum instructions. Follow those steps to locate your key when needed.
- You can find detailed instructions here: Vocareum_GenAI_API_Students_Guide.pdfDownload Vocareum_GenAI_API_Students_Guide.pdf

Security warning: Keep your API key *private*. Do not paste it in screenshots, chats, notebooks you share, or public repos. Prefer environment variables or provided key-management cells; never hard-code keys in source files. If you suspect exposure, rotate the key immediately per the guide.

To access Vocareum:

- Go to the *Getting Started* module on your Course homepage (just below Programme Orientation).

- Click the Vocareum link (labelled *Vocareum: Professional Certificate Programme in Agentic AI and Applications*).
- Alternatively, you can also access it from the Course menu on the left side of your screen.

Note: *Working through problems independently is key to building confidence and developing a lasting understanding. This is a self-study activity for your practice and does not count towards programme completion. No submission is required. However, we highly recommend completing the activity on Vocareum to enhance your overall learning experience.*